

$$\ln x \leq x - 1.$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots, \quad x \in \mathbb{R}.$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, \quad |x| < 1.$$

Markov's inequality: For nonnegative X ,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}, \quad a > 0.$$

Chebyshev's inequality:

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq a) \leq \frac{\text{Var}(X)}{a^2}, \quad a > 0.$$

Jensen's inequality: For convex f ,

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

1 Measures

1.1 Information

The **information content** of an event of probability p is uniquely (up to base) determined by **non-negativity**, $\psi(1) = 0$, **continuity**, **monotonicity**, and **additivity under independence**:

$$\psi(p) = \log_b \frac{1}{p}, \quad p \in [0, 1].$$

1.2 Shannon Entropy

Uniquely characterised by **continuity**, **maximality at uniform** (increasing in N), and the **successive-decisions rule**:

$$H(P_X) = \mathbb{E}_{X \sim P_X} \left[\log_2 \frac{1}{P_X(X)} \right].$$

Binary $H_2(p) = -p \log_2 p - (1-p) \log_2 (1-p)$;

Uniform over alphabet $\mathcal{X}$: $H = \log_2 |\mathcal{X}|$.

Joint/conditional:

$$H(X, Y) = \mathbb{E} \left[\log_2 \frac{1}{P_{X,Y}(X,Y)} \right],$$

$$H(Y|X) = \mathbb{E} \left[\log_2 \frac{1}{P_{Y|X}(Y|X)} \right].$$

$H(Y|X = x)$ may exceed $H(Y)$ for a particular x .

Properties of Shannon entropy

- $0 \leq H(X) \leq \log_2 |\mathcal{X}|$, eq. iff deterministic / uniform.
- Chain rule**: $H(X_1, \dots, X_n) = \sum_i H(X_i | X_1, \dots, X_{i-1})$.
- $H(X|Y) \leq H(X)$, eq. iff indep. ($H(X|Y = y)$ may exceed $H(X)$).
- Sub-additivity**: $H(X_1, \dots, X_n) \leq \sum_i H(X_i)$, eq. iff indep.
- Data processing inequality (DPI)**: $H(X) \geq H(f(X))$; eq. for bijection.
- Concavity**: $H(\lambda P + (1-\lambda)Q) \geq \lambda H(P) + (1-\lambda)H(Q)$.

1.3 KL Divergence

KL divergence between P, Q :

$$D_{\text{KL}}(P||Q) = \mathbb{E}_{X \sim P} \left[\log_2 \frac{P(X)}{Q(X)} \right].$$

- Gibbs**: $D \geq 0$, eq. iff $P = Q$; not symmetric, no triangle inequality.
- Chain rule**: $D(P_{XY}||Q_{XY}) = D(P_X||Q_X) + \sum_x P_X(x) D(P_{Y|x}||Q_{Y|x})$.
- DPI**: $D(P_X||Q_X) \geq D(P_{f(X)}||Q_{f(X)})$.

1.4 Mutual Information

Mutual information:

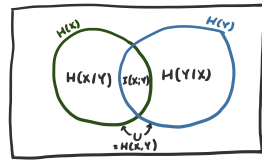
$$I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X).$$

$$\text{Conditional: } I(X; Y|Z) = H(X|Z) - H(X|Y, Z).$$

$$\text{Joint: } I(X; Y, Z) = H(X) - H(X|Y, Z).$$

$$I(X; Y) = \mathbb{E} \left[\log_2 \frac{P_{XY}(X,Y)}{P_X(X)P_Y(Y)} \right] = \mathbb{E} \left[\log_2 \frac{P_{Y|X}(Y|X)}{P_Y(Y)} \right].$$

Inclusion-exclusion: $I(X; Y, Z) = I(X; Y) + I(X; Z|Y)$.



Properties of mutual information

- Non-negativity**: $I(X; Y) \geq 0$, eq. iff indep.
- $I(X; Y) \leq H(X) \leq \log_2 |\mathcal{X}|$.
- $I(X; Y) \leq H(Y) \leq \log_2 |\mathcal{Y}|$.
- $I(X; Y) = H(X)$ iff X is a deterministic function of Y .
- Symmetry**: $I(X; Y) = I(Y; X)$.
- Chain rule**: $I(X_1, \dots, X_n; Y) = \sum_i I(X_i; Y | X_1, \dots, X_{i-1})$.
- DPI**: $X \rightarrow Y \rightarrow Z \Rightarrow I(X; Z) \leq I(X; Y), I(Y; Z)$; $W \rightarrow X \rightarrow Y \rightarrow Z \Rightarrow I(W; Z) \leq I(X; Y)$. Eq. iff $X \rightarrow Z \rightarrow Y$ (Z sufficient).
- Info-preserving**: Y depends on X through $f(X) \Rightarrow I(X; Y) = I(f(X); Y)$.
- Partial sub-add.**: Y_i cond. indep. given $\mathbf{X}$, each Y_i depends only on $X_i \Rightarrow I(\mathbf{X}; \mathbf{Y}) \leq \sum_i I(X_i; Y_i)$.
- Convexity**: convex in $P_{Y|X}$ (fixed P_X), concave in P_X (fixed $P_{Y|X}$).
- Conditioning can increase MI**: X, Y indep. Bernoulli, $Z = X \oplus Y$.

2 Symbol-wise Source Coding

Setup

We denote the set of all symbols (the **alphabet**) by $\mathcal{X}$. Consider a **discrete memoryless source (DMS)** $X \sim P_X$ over finite $\mathcal{X}$ (blocks: $P_{\mathbf{X}}(\mathbf{x}) = \prod_i P_X(x_i)$).

We encode each string $x \in \mathcal{X}^*$ into a decodeable binary string $C(x)$, of length $\ell(x)$. The goal is to minimize the **expected code-word length** of the code C :

$$L(C) = \mathbb{E}_{X \sim P_X} [\ell(X)] = \sum_{x \in \mathcal{X}} P_X(x) \cdot \ell(x),$$

2.1 Symbol-Wise Coding

Symbol-wise coding encodes **each symbol** $x \in \mathcal{X}$ into a code-word $C(x)$, and **encodes a string** $x_1 \dots x_n$ by concatenating the codewords $C(x_1) \dots C(x_n)$.

Decodeability Conditions

Non-singular codes: A code is non-singular if all codewords are distinct.

$$\forall x, x' \in \mathcal{X}, x \neq x' \implies C(x) \neq C(x').$$

Uniquely decodeable codes: A code is uniquely decodeable if any concatenation of codewords can be uniquely parsed into codewords. That is, for distinct $x_1 \dots x_n, x'_1 \dots x'_m \in \mathcal{X}^*$, $C(x_1) \dots C(x_n) \neq C(x'_1) \dots C(x'_m)$.

Uniquely decodeability is **non-trivial to verify**.

Prefix-free codes: A code is prefix-free if no codeword is a prefix of another codeword.

C is prefix-free $\implies C$ is uniquely decodeable.

Additionally, prefix-free codes can be decoded **instantaneously**, without needing to wait for the entire string to be received.

Kraft's inequality

Kraft's inequality states that for any prefix-free code C ,

$$\sum_{x \in \mathcal{X}} 2^{-\ell(x)} \leq 1.$$

Any prefix-free code can be **represented as a binary tree**, where each codeword corresponds to a leaf. The depth of the leaf is the codeword length. It follows, by random traversal of the tree, that the **probability of reaching a leaf** is $2^{-\ell(x)}$.

Existence (converse): for any lengths $\{\ell(x)\}_{x \in \mathcal{X}}$ satisfying Kraft's inequality, there exists a prefix-free code with these codeword lengths.

McMillan's extension: the inequality $\sum_x 2^{-\ell(x)} \leq 1$ also holds for uniquely decodeable codes, so **prefix-free codes lose nothing in $L(C)$** .

Entropy lower bound

Entropy provides a fundamental **limit on the expected codeword length** of any uniquely decodeable code:

$$L(C) \geq H(X),$$

with equality iff $P_X(x) = 2^{-\ell(x)}$ for all $x \in \mathcal{X}$. **Proof sketch**: let $Z = \sum_x 2^{-\ell(x)} \leq 1$ (Kraft/McMillan) and $Q_X(x) = 2^{-\ell(x)}/Z$; then $L(C) - H(X) = \log_2 \frac{1}{Z} + D(P_X||Q_X) \geq 0$.

Using the entropy lower bound, **Shannon-Fano coding** is a simple method to construct a **prefix-free code for a given distribution** P_X by rounding up the ideal length:

$$\ell(x) = \left\lceil \log_2 \frac{1}{P_X(x)} \right\rceil.$$

These lengths satisfy Kraft (since $\sum_x 2^{-\lceil \log_2 1/P_X(x) \rceil} \leq \sum_x P_X(x) = 1$), so a prefix-free code with these lengths exists. A Shannon-Fano coding satisfies the **expected codeword length** of

$$H(X) \leq L(C_{\text{SF}}) < H(X) + 1.$$

The +1 penalty can be significant when $H(X)$ is small (e.g. a source with $H(X) = 0.5$ bits still needs $L(C) \geq 1$ for any non-trivial code).

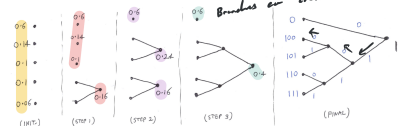
Mismatched coding: lengths $\ell(x) = \lceil \log_2 1/Q_X(x) \rceil$ give $H(X) + D(P_X||Q_X) \leq L(C_Q) < H(X) + D(P_X||Q_X) +$

wrong distribution costs $D(P_X||Q_X)$ extra bits.

2.2 Huffman Coding

Huffman coding constructs an **optimal prefix-free code** for a given distribution P_X . The **algorithm** is as follows:

- Start with a forest of $|\mathcal{X}|$ trees, each consisting of a single node corresponding to a symbol $x \in \mathcal{X}$, with weight $P_X(x)$.
 - While there are more than one tree in the forest:
 - Select the two roots with the smallest weights, say T_1 and T_2 .
 - Merge T_1 and T_2 into a new tree T , where T_1 and T_2 are the left and right subtrees of T , respectively. The weight of T is the sum of the weights of T_1 and T_2 .
 - The remaining tree in the forest is the Huffman tree.
- At each node, assign a binary digit to each branch. The code for a given symbol is the **sequence of binary digits along the path** from the root to the leaf corresponding to that symbol.



Optimality: Huffman achieves the minimum $L(C)$ over all uniquely decodeable codes:

$$L(C_H) = \min_{C: \text{UD}} L(C), \quad H(X) \leq L(C_H) < H(X) + 1.$$

Proof sketch (exchange argument): any optimal prefix-free code satisfies the invariants maintained by Huffman's merge step.

2.3 Extensions of Symbol-wise Coding

Block coding: Instead of encoding each symbol separately, we can **encode blocks of n symbols** together. We consider instead the **alphabet $\mathcal{X}^n$** , and the distribution P_{X^n} .

This allows us to achieve expected codeword lengths closer to the entropy, as n increases. $\forall \epsilon > 0$, there exists a block code C s.t.:

$$L(C) < H(X) + \epsilon.$$

Advantages of block coding:

- Encode sources with memory, by treating blocks as symbols. **P(eapple) > P(ealpp)** despite containing the same characters.
- Achieve expected codeword length arbitrarily close to the entropy. Even with independent symbols,

$$NH(X) \leq L(C) \leq NH(X) + 1.$$

$$H(X) \leq \text{Average length per symbol} \leq H(X) + \frac{1}{N}.$$

Disadvantages of block coding:

- Determining the product distribution P_{X^n} can be difficult, especially for sources with memory.
- Encoding and decoding complexity grows exponentially with n . Requires sorting $|\mathcal{X}|^n$ symbols by probability, and storing the codebook of size $|\mathcal{X}|^n$.

3 Block Source Coding

Setup

Fixed-to-fixed length source coding: take a length- n block $\mathbf{X} = (X_1, \dots, X_n) \in \mathcal{X}^n$ from a **DMS**, apply

- Encoder** $f: \mathcal{X}^n \rightarrow \{1, \dots, M\}$, **output** $m = f(\mathbf{X})$;
 - Decoder** $g: \{1, \dots, M\} \rightarrow \mathcal{X}^n$, **estimate** $\hat{\mathbf{X}} = g(m)$.
- If $\log_2 M < n \log_2 |\mathcal{X}|$ some sequences *must* be decoded wrongly: unavoidable errors, unlike variable-length coding. **Error probability** $P_e = \mathbb{P}[\hat{\mathbf{X}} \neq \mathbf{X}]$; **Rate** $R = \frac{1}{n} \log_2 M$ (bits per source symbol).

3.1 Fixed-Length Source Coding Theorem

For a DMS with per-symbol entropy $H(X)$:

- Achievability**: if $R > H(X)$, then $\forall \epsilon > 0$ there exist (sufficiently large) n , encoder f , decoder g of rate R with $P_e \leq \epsilon$.
- Converse**: if $R < H(X)$, then $\exists \epsilon > 0$ such that *every* code of rate R has $P_e > \epsilon$ (for all n).
- Strong converse**: $P_e \rightarrow 1$ as $n \rightarrow \infty$ when $R < H(X)$.

3.2 Typical Set & AEP

For some fixed small constant $\epsilon > 0$, the **typical set** is

$$\mathcal{T}_n(\epsilon) = \left\{ \mathbf{x} \in \mathcal{X}^n : \left| \frac{1}{n} \sum_{i=1}^n \log_2 \frac{1}{P_X(x_i)} - H(X) \right| \leq \epsilon \right\}$$

Typical set properties

- Probability bounds**: $2^{-n(H(X)+\epsilon)} \leq P_{\mathbf{X}}(\mathbf{x}) \leq 2^{-n(H(X)-\epsilon)}$ for $\mathbf{x} \in \mathcal{T}_n(\epsilon)$.
- High probability**: $\mathbb{P}[\mathbf{X} \in \mathcal{T}_n(\epsilon)] \rightarrow 1$ as $n \rightarrow \infty$ (LLN on i.i.d. $\log_2 \frac{1}{P_X(X_i)}$, mean $H(X)$).
- Cardinality upper bound**: $|\mathcal{T}_n(\epsilon)| \leq 2^{n(H(X)+\epsilon)}$ (since $1 \geq \sum_{\mathbf{x} \in \mathcal{T}_n} P_{\mathbf{X}}(\mathbf{x}) \geq |\mathcal{T}_n| \cdot 2^{-n(H(X)+\epsilon)}$).
- Cardinality lower bound**: $|\mathcal{T}_n(\epsilon)| \geq (1 - o(1)) 2^{n(H(X)-\epsilon)}$.

Asymptotic Equipartition Property (AEP): with high probability (w.h.p.) $\mathbf{X}$ is one of $\approx 2^{nH(X)}$ nearly-equiprobable sequences; $P_{\mathbf{X}}(\mathbf{X})^{1/n} \rightarrow 2^{-H(X)}$, and $(\prod_i f(X_i))^{1/n} \rightarrow 2^{\mathbb{E}[\log_2 f(X)]}$. **Ex (biased p)**: $\mathbf{x}$ with m ones typical iff $|\frac{m}{n} - p| \log_2 \frac{1-p}{p} \leq \epsilon$; $p = \frac{1}{2}$: all 2^n strings typical.

Strong typicality

$\mathcal{T}_n^s(\varepsilon) = \{\mathbf{x} : nP_X(x)(1-\varepsilon) \leq n_x(\mathbf{x}) \leq nP_X(x)(1+\varepsilon) \forall x\}$. **Lemma:** $\mathbf{x} \in \mathcal{T}_n^s(\varepsilon) \Rightarrow (1-\varepsilon)\mathbb{E}[a(X)] \leq \frac{1}{n} \sum_i a(x_i) \leq (1+\varepsilon)\mathbb{E}[a(X)]$ for $a \geq 0$; $a = \log_2 \frac{1}{P_X}$ recovers weak typicality.

Achievability via typicality

Assign a distinct index in $\{1, \dots, M-1\}$ to each $\mathbf{x} \in \mathcal{T}_n(\varepsilon)$, reserve M as a "dummy" (error) for $\mathbf{x} \notin \mathcal{T}_n(\varepsilon)$.

$$M = |\mathcal{T}_n(\varepsilon)| + 1 \leq 2^{n(H(X)+\varepsilon)} + 1, \\ R = \frac{1}{n} \log_2 M \rightarrow H(X) + \varepsilon; P_e \rightarrow 0.$$

3.3 Fano's Inequality

For discrete $X, \hat{X}$ on a common finite alphabet $\mathcal{X}$ with $P_e = \mathbb{P}[\hat{X} \neq X]$,

$$H(X | \hat{X}) \leq H_2(P_e) + P_e \log_2(|\mathcal{X}| - 1)$$

where $H_2(\alpha) = \alpha \log_2 \frac{1}{\alpha} + (1-\alpha) \log_2 \frac{1}{1-\alpha}$.

Intuition: first ask "is $\hat{X} = X$?" (cost $H_2(P_e)$); if not, X lies in the remaining $|\mathcal{X}| - 1$ values (cost $\leq \log_2(|\mathcal{X}| - 1)$, weighted by P_e).

Proof sketch: With $E = \mathbf{1}\{X \neq \hat{X}\}$,

$$H(X | \hat{X}) = H(X, E | \hat{X}) = H(E | \hat{X}) + H(X | \hat{X}, E) \\ \leq H(E) + P_e \log_2(|\mathcal{X}| - 1).$$

Weaker form (using $H_2(P_e) \leq 1$ and $|\mathcal{X}| - 1 \leq |\mathcal{X}|$):

$$H(X | \hat{X}) \leq 1 + P_e \log_2 |\mathcal{X}|.$$

Equality example: e.g. $\hat{X}$ uniform on $\mathcal{X}$ with $P(X = \hat{x} | \hat{X} = \hat{x}) = 1 - P_e$ and the other $|\mathcal{X}| - 1$ values equiprobable.

3.4 Fano Converse

Apply weaker Fano with $(X, \hat{X}) \leftarrow (\mathbf{X}, \hat{\mathbf{X}})$ on alphabet $\mathcal{X}^n$:

$$H(\mathbf{X} | \hat{\mathbf{X}}) \leq 1 + P_e \cdot n \log_2 |\mathcal{X}|.$$

$$nR \geq H(\hat{\mathbf{X}}) \\ \geq I(\mathbf{X}; \hat{\mathbf{X}}) \\ \geq nH(X) - P_e \cdot n \log_2 |\mathcal{X}| - 1. \\ \therefore P_e \geq \frac{1}{\log_2 |\mathcal{X}|} \left(H(X) - R - \frac{1}{n} \right).$$

So $R < H(X) \Rightarrow P_e \not\rightarrow 0$.

4 Channel Coding

Discrete memoryless channel (DMC): triple $(\mathcal{X}, P_{Y|X}, \mathcal{Y})$; memoryless over n uses: $P_{\mathbf{Y}|\mathbf{X}}(\mathbf{y}|\mathbf{x}) = \prod_i P_{Y_i|X_i}(y_i|x_i)$.

Channel matrix $Q_{j,i} = P_{Y|X}(j|i)$, so $P_Y = Q P_X$.

Setup

Uniform **Message** $m \in \{1, \dots, M\}$; **Encoder** sends $\mathbf{x}^{(m)} \in \mathcal{X}^n$ from **Codebook** $\mathcal{C}$; **Decoder** $g : \mathcal{Y}^n \rightarrow \{1, \dots, M\}$ outputs $\hat{m}$.

Error $P_e = \mathbb{P}[\hat{m} \neq m]$; **Rate** $R = \frac{1}{n} \log_2 M$ so $M = 2^{nR}$.

4.1 Channel Capacity

Capacity C = sup of rates R such that

$\forall \varepsilon > 0, \exists n, \mathcal{C}$ with $M = 2^{nR}$ and $P_e \leq \varepsilon$.

Channel Coding Theorem

$$C = \max_{P_X} I(X; Y)$$

Any maximiser is a **capacity-achieving input distribution**.

- Achievability:** $R < C \Rightarrow \exists$ codes with $P_e \rightarrow 0$.
- Converse:** $R > C \Rightarrow P_e$ bounded away from 0.

Alphabet cap: $C \leq \min\{\log_2 |\mathcal{X}|, \log_2 |\mathcal{Y}|\}$; binary input $\Rightarrow C \leq 1$ regardless of $|\mathcal{Y}|$.

4.2 Example Channels

- Noiseless,** $Y = X$: $C = \log_2 |\mathcal{X}|$.
- Binary Symmetric Channel** (BSC)(δ): flip with probability δ . Uniform $P_X \Rightarrow$ Uniform $P_Y \Rightarrow H(Y) = 1$.
 $C = 1 - H_2(\delta), P_X(0) = P_X(1) = \frac{1}{2}$.
- Binary Erasure Channel** (BEC)(ε): $\mathcal{Y} = \{0, 1, e\}$, erase with probability ε . $H(X|Y=0) = H(X|Y=1) = 0$, $H(X|Y=e) = H(X)$
 $\Rightarrow I(X; Y) = H(X) - H(X|Y) = (1-\varepsilon)H(X)$.
 $C = 1 - \varepsilon, P_X$ Uniform.
- Symmetric channels** (rows/cols of Q are permutations of each other): $H(Y|X)$ constant in x , Uniform P_X is capacity-achieving, and $C = \log_2 |\mathcal{Y}| - H(\text{row of } Q)$.
- Mod-sum:** $Y = (X + Z) \bmod m, Z \in \{0, 1\}$ Bernoulli(α). Uniform $P_X \Rightarrow C = \log_2 m - H_2(\alpha)$.
- Cascade BSC(δ) $\rightarrow$ BEC(ε):** let $E = \mathbf{1}\{Y=?\}$, decompose $H(Y|X) = H(E|X) + H(Y|E, X)$; $C = (1-\varepsilon)(1-H_2(\delta))$.
- Non-symmetric:** optimal P_X may be non-uniform; exploit symmetry to reduce P_X to few parameters (e.g. $P_X = (1-2p, p, p)$), set $\partial I / \partial p = 0$ (Lagrangian / KKT).
- Z-channel** ($P(0|0)=1, P(1|1)=1-q$): $C_Z \geq \frac{1-q}{2}$ by emulating erasure.

4.3 Jointly Typical Set

For joint distribution P_{XY} , fix $\varepsilon > 0$:

$$\mathcal{T}_n(\varepsilon) = \{(\mathbf{x}, \mathbf{y}) : \text{all three typical}\}$$

i.e. $\mathbf{x}$ P_X -typical, $\mathbf{y}$ P_Y -typical, and $(\mathbf{x}, \mathbf{y})$ P_{XY} -typical:

$$2^{-n(H(X,Y)+\varepsilon)} \leq P_{\mathbf{X}\mathbf{Y}}(\mathbf{x}, \mathbf{y}) \leq 2^{-n(H(X,Y)-\varepsilon)},$$

and analogously for the X - and Y -marginal bounds.

Four properties

- Equivalent defn:** empirical log-likelihoods $\frac{1}{n} \sum \log_2 \frac{1}{P(\cdot)}$ within ε of $H(X), H(Y), H(X, Y)$.
- High prob:** $\mathbb{P}[(\mathbf{X}, \mathbf{Y}) \in \mathcal{T}_n(\varepsilon)] \rightarrow 1$ (LLN).
- Cardinality:** $|\mathcal{T}_n(\varepsilon)| \leq 2^{n(H(X,Y)+\varepsilon)}$.
- Mismatched:** if $\mathbf{Z} \stackrel{\text{iid}}{\sim} Q_Z$, then $\mathbb{P}[\mathbf{Z} \in \mathcal{T}_n(\varepsilon; P_Z)] \approx 2^{-nD(P_Z \| Q_Z)}$.
- Independent pair:** if $(\mathbf{X}', \mathbf{Y}') \sim P_{\mathbf{X}} P_{\mathbf{Y}}$ (independent), $\mathbb{P}[(\mathbf{X}', \mathbf{Y}') \in \mathcal{T}_n(\varepsilon)] \leq 2^{-n(I(X;Y)-3\varepsilon)}$.

4.4 Achievability (Random Coding)

Codebook

Draw each symbol $X_i^{(m)} \stackrel{\text{iid}}{\sim} P_X$ independently for all i, m .

Decoder

Output $\hat{m}$ if it is the *unique* index with $(\mathbf{x}^{(\hat{m})}, \mathbf{Y}) \in \mathcal{T}_n(\varepsilon)$; else error.

Error analysis

Averaged $\overline{P_e^{(m)}} \leq \delta_n + M \cdot 2^{-n(I(X;Y)-3\varepsilon)}$. With $M = 2^{nR}$, second term $\rightarrow 0$ when $R < I(X; Y) - 3\varepsilon$. Pick P_X achieving C ; deterministic code exists by probabilistic method.

4.5 Converse via Fano

Weak Fano: $H(m|\hat{m}) \leq 1 + P_e \log_2 M$, so $nR = H(m) = I(m; \hat{m}) + H(m|\hat{m}) \leq I(m; \hat{m}) + 1 + P_e nR$. Along

$$m \rightarrow \mathbf{X} \rightarrow \mathbf{Y} \rightarrow \hat{m}: I(m; \hat{m}) \stackrel{\text{DPI}}{\leq} I(\mathbf{X}; \mathbf{Y}) \leq$$

$\sum_i I(X_i; Y_i) \leq nC$ (sub-additivity + memorylessness). Thus $P_e \leq 1 - \frac{C+1/n}{R} > 0$ when $R > C$.

4.6 Joint Source-Channel Coding

Source of entropy rate H , k symbols over DMC, n uses. Reliable iff $nC > kH \iff \frac{k}{n} < \frac{C}{H}$. **Separation theorem:** source-compress to $\approx 2^{kH}$ bits then channel-code is asymptotically optimal.

5 Continuous RVs & Gaussian Channel

5.1 Differential Entropy

Differential entropy: for continuous X with PDF f_X ,

$$h(X) = \mathbb{E}_{f_X} \left[\log_2 \frac{1}{f_X(X)} \right] = - \int_{\mathbb{R}} f_X(x) \log_2 f_X(x) dx.$$

Joint and Conditional versions:

$$h(X, Y) = \mathbb{E} \left[\log_2 \frac{1}{f_{XY}(X, Y)} \right],$$

$$h(Y|X) = \int f_X(x) h(Y|X=x) dx.$$

Properties that still hold

- Chain rule:** $h(X_1, \dots, X_n) = \sum_i h(X_i | X_1, \dots, X_{i-1})$.
- Conditioning reduces:** $h(X|Y) \leq h(X)$.
- Sub-additivity:** $h(X_1, \dots, X_n) \leq \sum_i h(X_i)$.
- Translation:** $h(X + c) = h(X)$.

Properties that fail

- Non-negativity:** h can be negative.
- Scaling:** $h(cX) = h(X) + \log_2 |c|$ (so not invariant under 1-1 maps); h not comparable to discrete H .

Standard forms

Uniform $[a, b]$: $h = \log_2(b - a)$.

Gaussian $\mathcal{N}(\mu, \sigma^2)$: $h = \frac{1}{2} \log_2(2\pi e \sigma^2)$.

Laplace(b): $h = \log_2(2be)$.

5.2 KL Divergence & Mutual Information

$$D(f \| g) = \int f(x) \log_2 \frac{f(x)}{g(x)} dx,$$

$$I(X; Y) = D(f_{XY} \| f_X f_Y) \\ = h(Y) - h(Y|X) = h(X) - h(X|Y).$$

- All discrete-case properties hold: **Non-negativity** ($D \geq 0, I \geq 0$), **Chain rule**, **DPI**.
- $I(X; Y) = I(\phi(X); \psi(Y))$ for invertible ϕ, ψ .

5.3 Gaussian Maximises Entropy

$h(X) \leq \frac{1}{2} \log_2(2\pi e \text{Var}[X])$, equality iff **Gaussian**.

Multivariate: $h(\mathbf{X}) \leq \frac{1}{2} \log_2 \det(2\pi e \text{Cov}[\mathbf{X}])$, tight for $\mathcal{N}(\mu, \Sigma)$.

5.4 Additive White Gaussian Noise (AWGN) Channel

$Y = X + Z, Z \sim \mathcal{N}(0, \sigma^2)$ independent of X . Without a constraint $C = \infty$. Consider $\mathbb{E}[X^2] \leq P$. Capacity $C(P) = \max_{f_X: \mathbb{E}[X^2] \leq P} I(X; Y)$:

$$C(P) = \frac{1}{2} \log_2 \left(1 + \frac{P}{\sigma^2} \right) \text{ bits/use}$$

achieved by $X \sim \mathcal{N}(0, P)$.

Derivation

$I(X; Y) = h(Y) - h(Z)$ (shift + indep.); $\text{Var}[Y] \leq P + \sigma^2$, so $h(Y) \leq \frac{1}{2} \log_2(2\pi e(P + \sigma^2))$, tight iff $X \sim \mathcal{N}(0, P)$.

Behaviour

Depends only on **SNR** = P/σ^2 .

Low SNR: $C \approx \frac{P}{2\sigma^2} \log_2 e$.

High SNR: $C \approx \frac{1}{2} \log_2(P/\sigma^2)$ (doubling P adds $\frac{1}{2}$ bit).

6 Codes

BSC codes: $\mathbf{u} \in \{0, 1\}^k \mapsto \mathbf{x} \in \{0, 1\}^n, \mathbf{y} = \mathbf{x} \oplus \mathbf{z}$, decoder outputs $\hat{\mathbf{u}}$. Rate $R = k/n$.

6.1 Linear Codes

Generator matrix

$\mathbf{x} = \mathbf{u}\mathbf{G}$ in mod-2 arithmetic, $\mathbf{G} \in \{0, 1\}^{k \times n}$.

- Systematic:** $\mathbf{G} = [\mathbf{I}_k \mathbf{P}]$; first k bits are $\mathbf{u}$, remaining $n - k$ are parity checks. Any general $\mathbf{G}$ reduces to systematic via row-swaps + mod-2 Gaussian elim.
- Linearity:** $\mathbf{x} \oplus \mathbf{x}'$ is also a codeword; $\mathbf{0} \in \mathcal{C}$. **Storage** nk bits (vs. $n2^{nR}$ random).

Examples

Repetition $(n, 1)$: $\mathbf{G} = [1 \dots 1]$, majority-vote; $R=1/n$, $d_{\min}=n$, corrects $\lfloor (n-1)/2 \rfloor$ flips. **Hamming order r :** $n=2^r-1, k=2^r-r-1, \mathbf{G} = [\mathbf{I}_k \mathbf{B}]$ with rows of $\mathbf{B}$ = all length- r vectors of weight ≥ 2 ; $d_{\min}=3$, corrects 1 flip; syndrome $\mathbf{yH}$ = binary address of erred bit. **(7, 4) instance:**

$$\mathbf{G}_{\text{Ham}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}, 16 \text{ codewords.}$$

Parity-check matrix

$\mathbf{H} \in \{0, 1\}^{n \times (n-k)}$: $\mathbf{xH} = \mathbf{0} \iff \mathbf{x} \in \mathcal{C}$. If $\mathbf{G} = [\mathbf{I}_k \mathbf{P}]$ then $\mathbf{H} = \begin{bmatrix} \mathbf{P} \\ \mathbf{I}_{n-k} \end{bmatrix}$. **H checks** without $O(2^k)$ enumeration.

6.2 Distance Properties

Hamming distance $d_H(\mathbf{x}, \mathbf{x}') = \#\{i : x_i \neq x'_i\}$, weight $w(\mathbf{x}) = \#\{i : x_i = 1\}$. Min distance $d_{\min} = \min_{\mathbf{x} \neq \mathbf{x}'} d_H$.

- Linear:** $d_{\min} = \min_{\mathbf{x} \neq \mathbf{0}} w(\mathbf{x})$.
- Correction:** up to $d_{\min} - 1$ erasures, $\lfloor (d_{\min} - 1)/2 \rfloor$ flips (disjoint Hamming balls).
- BSC success** (t -flip-corr., $t = \lfloor (d_{\min} - 1)/2 \rfloor$): $P_{\text{succ}} \geq \sum_{i=0}^t \binom{n}{i} \delta^i (1-\delta)^{n-i}$; $t=1: (1-\delta)^n + n\delta(1-\delta)^{n-1}$.
- Hamming (7, 4):** $d_{\min} = 3$, corrects 1 flip.

6.3 Minimum-Distance Decoding

ML = min-distance (BSC)

For uniform m , ML ($\arg \max_j P(\mathbf{y}|\mathbf{x}^{(j)})$) minimises P_e . On BSC($\delta < \frac{1}{2}$): $\log P(\mathbf{y}|\mathbf{x}) = n \log(1 - \delta) - d_H(\mathbf{x}, \mathbf{y}) \log \frac{1-\delta}{\delta}$, so ML = $\arg \min_j d_H(\mathbf{x}^{(j)}, \mathbf{y})$.

Syndrome

$\mathbf{S} = \mathbf{yH} = \mathbf{zH}$ (depends only on error).

- Decode: $\hat{\mathbf{z}} = \arg \min_{\hat{\mathbf{z}}\mathbf{H}=\mathbf{S}} w(\hat{\mathbf{z}}), \hat{\mathbf{x}} = \mathbf{y} \oplus \hat{\mathbf{z}}$.
- Algorithm:** $\mathbf{S} = \mathbf{0} \Rightarrow \hat{\mathbf{x}} = \mathbf{y}$; else try $\hat{\mathbf{z}}$ of weight 1, 2, ... until $\hat{\mathbf{z}}\mathbf{H} = \mathbf{S}$.
- Hamming (7, 4):** rows of $\mathbf{H}$ are the 7 nonzero length-3 vectors; each single-bit weight has a unique syndrome $\Rightarrow$ direct lookup.

Hamming bound

A (n, k) code correcting all weight- $\leq t$ errors needs $\sum_{i=0}^t \binom{n}{i} \leq 2^{n-k}$ (distinct syndromes).